



COURSE PACK

Course pack and syllabus

Incident Response Fundamentals

An 8-week course from reading one EDR alert to running a ransomware incident through every NIST 800-61 phase.

Length 8 weeks

Discipline Incident Response

Level Upper undergraduate, graduate, or professional cohort

Prerequisites Networking and operating systems fundamentals. Introduction to Digital Forensics is helpful but not required.

Prepared 2026-10-07

Every lab in this course generates a different variant for each student from one case, with an answer key verified against that student's own evidence. Grading and the student portal run on the instructor's machine.

Learning outcomes

1. Triage EDR, SIEM and proxy evidence by severity and ATT&CK technique.
2. Scope an intrusion from initial access to impact.
3. Choose and justify containment, eradication and recovery actions.
4. Produce indicators and a breach report a manager can act on.

Assessment

Weekly labs 50% (best 6 of 7), timed drills 30%, final incident 20%.

Tools

Any JSON viewer, jq, Excel or LibreOffice, Wireshark, EvtxECmd, Timeline Explorer. All free; one-run install scripts are on the Tools page.

Week by week

Wk	Topic	Objectives	Case and mode	Assessment
1	Reading an EDR alert	Triage by severity and read a process tree. Map a detection to an ATT&CK technique.	First Look: Reading an EDR Alert (practice)	Lab 1
2	The SIEM queue	Read a SIEM alert export. Separate a true positive from noise.	First Look: Reading a SIEM Alert (practice)	Lab 2
3	Beacons in web traffic	Spot periodic egress in proxy logs. Pull a C2 indicator set.	First Look: One Beacon in the Web Traffic (practice)	Lab 3
4	Phishing to foothold	Follow a phish through delivery, execution and persistence. Make the first containment call under time.	Phishing to Foothold (timed exam, 45 min)	Drill 1 (15%)
5	Malware triage	Go from alert to verdict with hashes, signatures and behaviour.	Malware Triage: From Alert to Verdict (practice)	Lab 4
6	Lateral movement	Pair 4624 logons with 7045 service installs. Scope which hosts the attacker reached.	Lateral Move to the File Server (practice)	Lab 5
7	Business email compromise	Trace a fraudulent wire request through mail and sign-in evidence. Decide what to revoke first.	Business Email Compromise: Wire Fraud (timed exam, 45 min)	Drill 2 (15%)

Wk	Topic	Objectives	Case and mode	Assessment
8	Final incident: ransomware	Run a live ransomware chain through detection, containment, eradication and recovery. Write the breach report.	Ransomware Before Coffee (timed exam, 90 min, report)	Final 20%

Running the course

In Red Herrings, open Courses, pick this course, enter the class size or roster and press Generate the whole course. One job per week is created with the mode and time limit above; hand out each week's zips or portal links when that week arrives. Scores export to your LMS from each job.